

HW #2 - David Alvarado

1) $\vec{\nabla} \vec{v} = ?$ $\rho \vec{\nabla} \vec{v}$

$$\begin{matrix} x=1 \\ y=2 \\ z=3 \end{matrix}$$

$\rho \vec{\nabla} \vec{v} = ?$

$\left[\begin{matrix} b \\ 10 \\ 10 \end{matrix} \right]$

a. $v_x = b_y$ $v_y = 0$ $v_z = 0$

$$\vec{v} = b_y \vec{e}_1 + 0 \vec{e}_2 + 0 \vec{e}_3$$

$$\vec{\nabla} \vec{v} = \left(\sum_i \frac{\partial}{\partial x_i} \vec{e}_i \right) \left(\sum_j v_j \vec{e}_j \right)$$

$$= \sum_i \sum_j \frac{\partial}{\partial x_i} v_j \vec{e}_i \vec{e}_j$$

$$= \frac{\partial}{\partial x_1} [b_y] \vec{e}_1 \vec{e}_1 + \frac{\partial}{\partial x_1} [0] \vec{e}_1 \vec{e}_2 + 0$$

$$+ \frac{\partial}{\partial x_2} [b_y] \vec{e}_2 \vec{e}_1 + 0 + 0$$

$$+ \frac{\partial}{\partial x_3} [b_y] \vec{e}_3 \vec{e}_1 + 0 + 0$$

$$\vec{\nabla} \vec{v} = \frac{\partial}{\partial x_1} [b_y] \vec{e}_1 \vec{e}_1$$

$$\vec{\nabla} \vec{v} = b_y \vec{e}_1 \vec{e}_1$$

$$\begin{bmatrix} 0 & 0 & 0 \\ b & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

b. $v_x = b_y$ $v_y = b_x$ $v_z = 0$

$$\vec{\nabla} \vec{v} = \sum_i \sum_j \frac{\partial}{\partial x_i} v_j \vec{e}_i \vec{e}_j$$

$$= \frac{\partial}{\partial x_1} [b_y] \vec{e}_1 \vec{e}_1 + \frac{\partial}{\partial x_1} [b_x] \vec{e}_1 \vec{e}_2 + \frac{\partial}{\partial x_1} [0] \vec{e}_1 \vec{e}_3$$

$$+ \frac{\partial}{\partial x_2} [b_y] \vec{e}_2 \vec{e}_1 + \frac{\partial}{\partial x_2} [b_x] \vec{e}_2 \vec{e}_2 + 0$$

$$+ \frac{\partial}{\partial x_3} [b_y] \vec{e}_3 \vec{e}_1 + 0 + 0$$

$$= b_y \vec{e}_1 \vec{e}_1 + b_x \vec{e}_1 \vec{e}_2$$

$$\Rightarrow \begin{bmatrix} 0 & b & 0 \\ b & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

c. $v_x = -b_y$ $v_y = b_x$ $v_z = 0$

$$\vec{\nabla} \vec{v} = \sum_i \sum_j \frac{\partial}{\partial x_i} v_j \vec{e}_i \vec{e}_j$$

$$= \frac{\partial}{\partial x_1} [-b_y] \vec{e}_1 \vec{e}_1 + \frac{\partial}{\partial x_1} [b_x] \vec{e}_1 \vec{e}_2 + \frac{\partial}{\partial x_1} [0] \vec{e}_1 \vec{e}_3$$

$$+ \frac{\partial}{\partial x_2} [-b_y] \vec{e}_2 \vec{e}_1 + \frac{\partial}{\partial x_2} [b_x] \vec{e}_2 \vec{e}_2 + 0$$

$$+ \frac{\partial}{\partial x_3} [-b_y] \vec{e}_3 \vec{e}_1 + \frac{\partial}{\partial x_3} [b_x] \vec{e}_3 \vec{e}_2 + 0$$

$$= -b_y \vec{e}_1 \vec{e}_1 + b_x \vec{e}_1 \vec{e}_2$$

$$\Rightarrow \begin{bmatrix} 0 & b & 0 \\ -b & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\vec{v} = b_y \vec{e}_1 + 0 \vec{e}_2 + 0 \vec{e}_3$$

$$\vec{\nabla} \vec{v} = \left(\sum_i v_i \vec{e}_i \right) \left(\sum_j v_j \vec{e}_j \right)$$

$$= \sum_i \sum_j v_i v_j \vec{e}_i \vec{e}_j$$

$$= (b_y)(b_y) \vec{e}_1 \vec{e}_1 + b_y(0) \vec{e}_1 \vec{e}_2 + 0$$

$$= 0(b_y) \vec{e}_1 \vec{e}_2 + 0 + 0$$

$$= 0(0) \vec{e}_1 \vec{e}_3 + 0 + 0$$

$$= (b^2 y^2 \vec{e}_1 \vec{e}_1)$$

$$\Rightarrow \begin{bmatrix} b^2 y^2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\rho \vec{\nabla} \vec{v} = \rho \left(\sum_i v_i \vec{e}_i \right) \left(\sum_j v_j \vec{e}_j \right)$$

$$= (b_y b_y) \vec{e}_1 \vec{e}_1 + (b_y b_x) \vec{e}_1 \vec{e}_2 + 0$$

$$+ (b_x b_y) \vec{e}_2 \vec{e}_1 + b_x b_x \vec{e}_2 \vec{e}_2 + 0$$

$$+ 0 \vec{e}_3 \vec{e}_1 + 0 + 0$$

$$= b_y^2 \vec{e}_1 \vec{e}_1 + b_y b_x \vec{e}_1 \vec{e}_2 + b_x^2 \vec{e}_2 \vec{e}_2$$

$$\Rightarrow \begin{bmatrix} \rho b^2 y^2 & \rho b^2 y x & 0 \\ \rho b^2 x y & \rho b^2 x^2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\rho \vec{\nabla} \vec{v} = \rho \left(\sum_i v_i \vec{e}_i \right) \left(\sum_j v_j \vec{e}_j \right)$$

$$= \rho \left((-b_y) \cdot (-b_y) \vec{e}_1 \vec{e}_1 + (-b_y) b_x \vec{e}_1 \vec{e}_2 + 0 \right)$$

$$+ (b_x \cdot (-b_y) \vec{e}_2 \vec{e}_1) + (b_x \cdot b_x) \vec{e}_2 \vec{e}_2 + 0$$

$$+ 0 \vec{e}_3 \vec{e}_1 + 0 + 0$$

$$= \rho \left(b^2 y^2 \vec{e}_1 \vec{e}_1 - b^2 y x \vec{e}_1 \vec{e}_2 - b^2 x y \vec{e}_2 \vec{e}_1 \right)$$

$$+ b^2 x^2 \vec{e}_2 \vec{e}_2$$

$$\Rightarrow \begin{bmatrix} \rho b^2 y^2 & -\rho b^2 y x & 0 \\ -\rho b^2 x y & \rho b^2 x^2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

d. $V_x = -0.5bx$ $V_y = -0.5by$ $V_z = bz$

$\vec{\nabla} \cdot \vec{V} = \sum_i \sum_j \frac{\partial}{\partial x_i} V_j \vec{e}_i \cdot \vec{e}_j$

$= \frac{\partial}{\partial x} [-0.5bx] \vec{e}_1 \cdot \vec{e}_1 + \frac{\partial}{\partial x} [-0.5by] \vec{e}_1 \cdot \vec{e}_2 + \frac{\partial}{\partial x} [bz] \vec{e}_1 \cdot \vec{e}_3$
 $+ \frac{\partial}{\partial y} [-0.5bx] \vec{e}_2 \cdot \vec{e}_1 + \frac{\partial}{\partial y} [-0.5by] \vec{e}_2 \cdot \vec{e}_2 + \frac{\partial}{\partial y} [bz] \vec{e}_2 \cdot \vec{e}_3$
 $+ \frac{\partial}{\partial z} [-0.5bx] \vec{e}_3 \cdot \vec{e}_1 + \frac{\partial}{\partial z} [-0.5by] \vec{e}_3 \cdot \vec{e}_2 + \frac{\partial}{\partial z} [bz] \vec{e}_3 \cdot \vec{e}_3$

$= -0.5b \vec{e}_1 \cdot \vec{e}_1 - 0.5b \vec{e}_2 \cdot \vec{e}_2 + b \vec{e}_3 \cdot \vec{e}_3$

$\Rightarrow \begin{bmatrix} -0.5b & 0 & 0 \\ 0 & -0.5b & 0 \\ 0 & 0 & b \end{bmatrix}$

$R = 1m$



$\vec{\omega} = 5 \text{ rad/s}$

$\mu = 30 \text{ Pa} \cdot \text{s}$

$\vec{\tau} = \vec{r} \times \vec{F}$
 $\tau = r F \sin \phi$
 $\phi = 90^\circ$
 $\tau = r F$
 $F = \tau / r$

L.Circ

$\Rightarrow F = \mu \frac{V}{Y} \cdot A$

$\Rightarrow \tau / r = \mu \frac{V}{Y} \cdot A$

$\tau = \mu \frac{V}{Y} \cdot A \text{ or}$

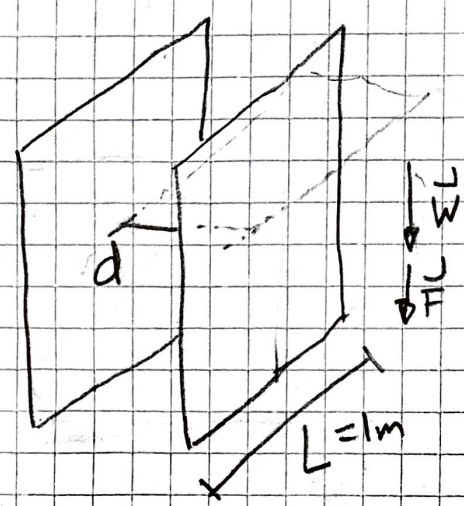
$\Rightarrow \tau = \mu \cdot \frac{r \omega}{Y} \cdot (L \cdot 2\pi r) \cdot r$

$= (30 \text{ Pa} \cdot \text{s}) \cdot \frac{(1m)(5 \text{ rad/s})}{0.005m} \cdot (1m \cdot 2\pi \cdot 1m) \cdot 1m$

$= 3289.868 \text{ Pa} \cdot \text{m}^3 \cdot \text{s} \cdot \frac{1}{\text{Pa}} \cdot \frac{\text{N/m}^2}{\text{Pa}}$

$\tau = 3289.868 \text{ N} \cdot \text{m}$

$= 3289.868 \text{ J}$



2)